owner in Japan who left this switch in the CTRL not required position. However, that person was quirky by nature.

The clock circuitry is mostly contained in the AY-5-3600. External components R1 and C5 set the operating frequency at approximately 80 KHz. Similarly C2 sets the keybounce mask period at 7 milliseconds.

The STROBE output works as it does in the MM5740 with a high level, one clock strobe output (12 microseconds) any time a matrix key is pressed. The REPEAT operation is different though. In the MM5740 a REPEAT input clock triggers the STROBE output if a matrix key is being held down. The AY-5-3600 is less sophisticated in this regard. It has an ANY KEY DOWN output which goes high when any key is held down. This is used in the Apple to enable a 15 Hz REPEAT oscillator when REPT and a matrix key are held. The REPEAT oscillator output is capacitively coupled to a circuit which generates a strobe of very roughly 45 microseconds when the REPEAT clock drops.

Capacitor C8 in the REPEAT oscillator circuit was apparently an afterthought. It is wired between pins 20 and 23 of PLUG 1 rather than being installed on the encoder board. It looks as though C8 is there to cause a short delay after REPT is held before the REPEAT oscillator is enabled.

The B1-B7 outputs of the II Plus encoder are routed to the keyboard/motherboard connector through inverters, just as on the old keyboard. There is a very important possibility incorporated, though. There are wired holes for mounting a slide switch which can select B9 and B8 instead of B5 and B6 as the K4 and K5 outputs. The B9 and B6 outputs are programmed for alphabetic shifting. This means that you can cut the two bow tie solder jumpers, install a slide switch, and have the option of enabling or disabling alphabetic shifting of the keyboard input. Table 7.2 shows the normal ASCII from the Apple keyboard and the ASCII from the II plus keyboard with alphabetic shifting enabled.

PERIPHERAL SLOT CONNECTIONS

It is hard to know where to start talking about peripheral slot I/O. You can do so much from the slots. They are as versatile as modern microcomputer architecture with full connection to the address bus and data bus. It's like someone designed a really neat computer but on the blueprints drew eight empty squares with the message, "user, please fill in the blanks." One never knows what lurks beneath the lid of an innocent looking Apple.

The capabilities of the peripheral slots seem more clear when you look at the connected signals in groups. Figure 7.9 illustrates the peripheral slot connections and groups the signals functionally.

Table 7.2 Relation of ASCII to Keypress. (1 of 2)

KEY	APPLE II SWITCH#	APPLE II Plus SWITCH#	NORMAL				APPLE II Plus LOWER CASE ENABLED			
			ALONE	CONTROL	SHIFT	BOTH	ALONE	CONTROL	SHIFT	BOTH
space	55	52	$A0	$A0	$A0	$A0	$A0	$A0	$A0	$A0
, <	50	48	$AC	$AC	$BC	$BC	$AC	$AC	$BC	$BC
- =	12	12	$AD	$AD	$BD	$BD	$AD	$AD	$BD	$BD
. >	51	49	$AE	$AE	$BE	$BE	$AE	$AE	$BE	$BE
/ ?	52	50	$AF	$AF	$BF	$BF	$AF	$AF	$BF	$BF
0	10	10	$B0	$B0	$B0	$B0	$B0	$B0	$B0	$B0
1 !	1	1	$B1	$B1	$A1	$A1	$B1	$B1	$A1	$A1
2 "	2	2	$B2	$B2	$A2	$A2	$B2	$B2	$A2	$A2
3 #	3	3	$B3	$B3	$A3	$A3	$B3	$B3	$A3	$A3
4 $	4	4	$B4	$B4	$A4	$A4	$B4	$B4	$A4	$A4
5 %	5	5	$B5	$B5	$A5	$A5	$B5	$B5	$A5	$A5
6 &	6	6	$B6	$B6	$A6	$A6	$B6	$B6	$A6	$A6
7 '	7	7	$B7	$B7	$A7	$A7	$B7	$B7	$A7	$A7
8 (	8	8	$B8	$B8	$A8	$A8	$B8	$B8	$A8	$A8
9)	9	9	$B9	$B9	$A9	$A9	$B9	$B9	$A9	$A9
: *	11	11	$BA	$BA	$AA	$AA	$BA	$BA	$AA	$AA
; +	38	37	$BB	$BB	$AB	$AB	$BB	$BB	$AB	$AB
A	29	28	$C1	$81	$C1	$81	$E1	$81	$C1	$81
B	47	45	$C2	$82	$C2	$82	$E2	$82	$C2	$82
C	45	43	$C3	$83	$C3	$83	$E3	$83	$C3	$83

Table 7.2 Relation of ASCII to Keypress. (2 of 2)

KEY	APPLE II SWITCH#	APPLE II Plus SWITCH#	NORMAL ALONE	NORMAL CONTROL	NORMAL SHIFT	NORMAL BOTH	APPLE II Plus LOWER CASE ENABLED ALONE	APPLE II Plus LOWER CASE ENABLED CONTROL	APPLE II Plus LOWER CASE ENABLED SHIFT	APPLE II Plus LOWER CASE ENABLED BOTH
D	31	30	$C4	$84	$C4	$84	$E4	$84	$C4	$84
E	17	17	$C5	$85	$C5	$85	$E5	$85	$C5	$85
F	32	31	$C6	$86	$C6	$86	$E6	$86	$C6	$86
G	33	32	$C7	$87	$C7	$87	$E7	$87	$C7	$87
H	34	33	$C8	$88	$C8	$88	$E8	$88	$C8	$88
l arrow	39	38	$88	$88	$88	$88	$88	$88	$88	$88
I	22	22	$C9	$89	$C9	$89	$E9	$89	$C9	$89
J	35	34	$CA	$8A	$CA	$8A	$EA	$8A	$CA	$8A
K	36	35	$CB	$8B	$CB	$8B	$EB	$8B	$CB	$8B
L	37	36	$CC	$8C	$CC	$8C	$EC	$8C	$CC	$8C
M]	49	47	$CD	$8D	$DD	$9D	$ED	$8D	$CD	$9D
RETURN	26	26	$8D	$8D	$8D	$8D	$8D	$8D	$8D	$8D
N ^	48	46	$CE	$8E	$DE	$9E	$EE	$8E	$CE	$9E
O	23	23	$CF	$8F	$CF	$8F	$EF	$8F	$CF	$8F
P @	24	24	$D0	$90	$C0	$80	$F0	$90	$D0	$80
Q	15	15	$D1	$91	$D1	$91	$F1	$91	$D1	$91
R	18	18	$D2	$92	$D2	$92	$F2	$92	$D2	$92
S	30	29	$D3	$93	$D3	$93	$F3	$93	$D3	$93
T	19	19	$D4	$94	$D4	$94	$F4	$94	$D4	$94
U	21	21	$D5	$95	$D5	$95	$F5	$95	$D5	$95
r arrow	40	39	$95	$95	$95	$95	$95	$95	$95	$95
V	46	44	$D6	$96	$D6	$96	$F6	$96	$D6	$96
W	16	16	$D7	$97	$D7	$97	$F7	$97	$D7	$97
X	44	42	$D8	$98	$D8	$98	$F8	$98	$D8	$98
Y	20	20	$D9	$99	$D9	$99	$F9	$99	$D9	$99
Z	43	41	$DA	$9A	$DA	$9A	$FA	$9A	$DA	$9A
ESC	14	14	$9B	$9B	$9B	$9B	$9B	$9B	$9B	$9B
RESET	13	13								
REPT	25	25								
CTRL	28	27								
SHIFT,l	42	40								
SHIFT,r	53	51								
II Plus KEYPAD										
0			$B0	$B0	$B0	$B0	$B0	$B0	$B0	$B0
1			$B1	$B1	$B1	$B1	$B1	$B1	$B1	$B1
2			$B2	$B2	$B2	$B2	$B2	$B2	$B2	$B2
3			$B3	$B3	$B3	$B3	$B3	$B3	$B3	$B3
4			$B4	$B4	$B4	$B4	$B4	$B4	$B4	$B4
5			$B5	$B5	$B5	$B5	$B5	$B5	$B5	$B5
6			$B6	$B6	$B6	$B6	$B6	$B6	$B6	$B6
7			$B7	$B7	$B7	$B7	$B7	$B7	$B7	$B7
8			$B8	$B8	$B8	$B8	$B8	$B8	$B8	$B8
9			$B9	$B9	$B9	$B9	$B9	$B9	$B9	$B9
space			$A0	$A0	$A0	$A0	$A0	$A0	$A0	$A0
+			$AB	$AB	$AB	$AB	$AB	$AB	$AB	$AB
,			$AC	$AC	$AC	$AC	$AC	$AC	$AC	$AC
-			$AD	$AD	$AD	$AD	$AD	$AD	$AD	$AD
.			$AE	$AE	$AE	$AE	$AE	$AE	$AE	$AE
RETURN			$8D	$8D	$8D	$8D	$8D	$8D	$8D	$8D